Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. **For each of the following matrices:**

   - **Obtain a row-equivalent matrix in echelon form using elementary row operations.**
   - **Determine its rank.**
   - **Obtain a row-equivalent matrix in reduced echelon form using elementary row operations.**

a) $B = \begin{bmatrix} 0 & 0 & 1 \\ 4 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$

i. **Row Echelon:**

$$\mathrm{B} \xrightarrow[t_1]{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & -1 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[t_2]{R_2 \leftarrow R_2 - 4R_1} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 4 \\ 0 & 0 & 1 \end{bmatrix} = B'$$

ii. **Rank:** $\operatorname{rank}(B) = 3$.

iii. **Reduced Echelon form:**

$$B' \xrightarrow[t_3]{R_2 \leftarrow -\frac{1}{7}R_2} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -\frac{4}{7} \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[t_4]{R_1 \leftarrow R_1 - 2R_2} \begin{bmatrix} 1 & 0 & \frac{1}{7} \\ 0 & 1 & -\frac{4}{7} \\ 0 & 0 & 1 \end{bmatrix}$$
$$\xrightarrow[t_5, t_6]{\substack{R_2 \leftarrow R_2 + \frac{4}{7}R_3 \\ R_1 \leftarrow R_1 - \frac{1}{7}R_3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = rref(B) = I_3$$

b)

$$C = \begin{bmatrix} 0 & 0 & 3 & 3i & -15 \\ i & 3 & 2 & 0 & 3 \\ 2 & -6i & -5i & 1 & 2 \end{bmatrix}$$

c)

$$D = \begin{bmatrix} 2 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \\ 4 & -4 & 4 & 4 \\ 0 & 2 & 0 & 0 \end{bmatrix}$$

d)

$$E = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

e)

$$F = \begin{bmatrix} 3 & 0 & 0 & -21 & 24 \\ 4 & 1 & 0 & -25 & 34 \\ -2 & 3 & 0 & 22 & -2 \end{bmatrix}$$